Retail Sales Analytics: Executive Summary

NexAfrica Data Analysis Internship, Task 1 | Superstore Sales, January 2014 to December 2017

Total Sales	Total Profit	Profit Margin	Orders	Customers	Avg Order Value	Units Sold
$2.30M	**$286.4K**	**12.5%**	**5,009**	**793**	**$458.56**	**37,871**

1. Business Problem

A retail company wants to understand how the business performs across products, customers, regions and time. The analysis answers: how much revenue and profit does the business generate, which products, categories, regions and customers perform best, how are sales changing over time, which products or areas need attention, and what should management do about it.

2. Dataset

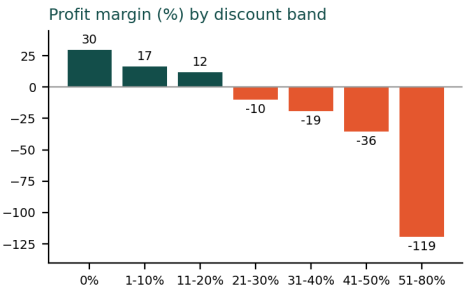
The Superstore Sales dataset (Kaggle) holds 9,994 order lines and 21 columns for January 2014 to December 2017. After cleaning, 9,993 lines remain, covering **793 customers, 5,009 orders, 1,850 products, 3 categories, 17 sub-categories and 4 regions**.

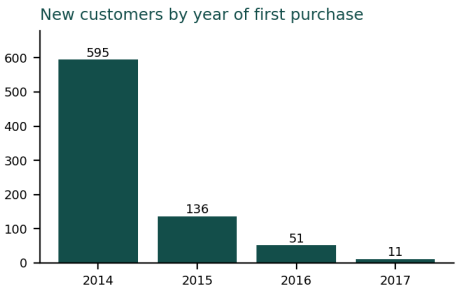
3. Methodology

- **Data cleaning:** standardized column names, parsed dates (month/day/year), restored postal codes that had lost leading zeros, trimmed text, checked missing values (none) and logic (no ship date before order date), and removed 1 exact duplicate. Negative profits and large orders were kept as genuine business events.
- **Transformation:** added year, month, year-month, shipping days, profit margin and a loss flag.
- **KPI and SQL analysis:** 7 core KPIs, sales trends, regions and categories, repeated in 13 SQL queries on a SQLite database (4 related tables) to cross-check the results.
- **Customer segmentation:** High-Value (top 25% by sales with positive profit), Low-Value (bottom 25% by sales) and Medium-Value (the rest).
- **Cohort analysis:** customers grouped by month of first purchase, with monthly and annual retention.
- **Discount and profitability analysis:** discount bands against profit, and products grouped by median sales and median profit.
- **Dashboard:** a 4-page interactive dashboard (Python, Streamlit and Plotly) with filters.

4. Key Findings

1. **Growth and seasonality.** Sales fell 2.8% in 2015, then grew 29.5% in 2016 and 20.4% in 2017. September, November and December bring in 43% of sales; January and February under 7%.
2. **Discounts above 20% destroy profit.** Margin falls from 29.5% with no discount to 11.6% at 11-20% and turns negative above 20%. The 1,392 lines discounted above 20% lost $135,364 (47% of total profit). Units per line did not rise with discount.
3. **Central over-discounts.** Its average discount is 24% (West 11%) and its margin is the weakest at 7.9% (West 14.9%). West leads in sales ($725K) and profit ($108K).
4. **Furniture sells but earns little.** It brings 32% of sales but 6% of profit (2.5% margin). Heavy discounts erase about 75% of its potential profit. Tables lose $17,725 overall.
5. **Technology and Office Supplies carry the profit,** together 94%. The Canon imageCLASS 2200 copier is the top product for both sales ($61.6K) and profit ($25.2K).
6. **252 high-selling products lose a combined $72,632.** 301 products (16%) lose money in total.
7. **Customer value is concentrated.** The top 20% of customers give 48% of sales and 59% of profit; the High-Value segment (22% of customers) gives 78% of profit. 155 customers (19.5%) are unprofitable.
8. **The customer base is loyal but not growing.** 73% to 87% of the 2014 customers are active in each later year, yet new customers fell from 595 in 2014 to 11 in 2017, and returning customers made 99% of 2017 sales.





5. Recommendations

1. **Cap discounts at 20% and require approval above it.** This targets the $135K lost on heavily discounted lines.
2. **Review discounting in the Central region,** starting with Binders, Tables and Machines.
3. **Reprice or review the high-sales, low-profit products:** raise prices, cut discounts or renegotiate supplier cost, and consider dropping the worst loss-makers.
4. **Review Furniture, especially Tables and Bookcases,** where margins are thin or negative.
5. **Restart customer acquisition and protect the core customers:** set a target for new customers and offer loyalty benefits to the High-Value segment.

6. Conclusion

The business is growing and profitable, with a 12.5% margin and a loyal customer base. Its main weakness is self-inflicted: discounting above 20% turns sales into losses, especially in the Central region and in Furniture, and a few high-selling products lose money. Fixing discount policy and pricing would raise profit without needing more sales. The second priority is to rebuild customer acquisition, because almost all recent growth comes from customers the business already had.

Limitations and possible improvements

The data comes from one retailer and ends in 2017. It has no cost, supplier or marketing data, so the causes of low margins are inferred from discounts and cannot be proven. Discounts are measured per order line, so their effect on total order counts is unknown. Because the data starts in January 2014, some 2014 'new' customers may have bought before. Customer segment thresholds are judgement-based. Possible improvements: add cost and marketing data, test discount caps on a sample of products, and forecast seasonal demand.